PROVEIT RESEARCH CONTINUATION

Mixed Hurwitz-Tail Moments, Zeta-Anchor Reductions, and Stieltjes Transport

Exact identities, a parity-refined finite compiler, and explicit rational certificates

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Abstract

We continue the harmonic-tail and Gamma–Stieltjes branches of the ProveIt polylogarithm programme. Three explicitly defined anchors in the existing fifth- and sixth-power moment generators are reduced to ordinary zeta products and, at weight eight, a single specified depth-two value. The reductions are proved by finite rational combinations of classical double-shuffle relations; complete short certificates are printed and independently replayed.

For an arbitrary product of integer-order Hurwitz tails, including odd and unequal orders, we give a terminating formula for every regular centered moment. The answer lies in the rational span of admissible multiple zeta values of weight at most $W-1$, where W is the sum of the tail orders. For odd centered degree the ceiling improves to $W-2$. A separate, simpler formula treats the full $(2,3)$ family and gives an ordinary polynomially subtracted series with value exactly -15; we supply a direct finite telescoper for it.

Finally, weighting by successive differences of a tail product gives all-index generalized-Stieltjes sums, polygamma identities, anchored Gamma-ratio sums, and primitives in terms of Hurwitz order derivatives at zero. A finite multiple-polylogarithm kernel identifies their coefficients with convergent final-index multiple-zeta derivatives. The article distinguishes these continuations from the already proved all-power even-tail recurrences. It includes source-status notes, an independently replayable exact certificate suite, numerical diagnostics, and a precise further-research agenda.

Scope. The claims are relative to the explicitly inspected source files, not an exhaustive priority claim over the literature or the repository. No arithmetic independence, Gaussian S_6/S_8 identity, or proof-assistant formalization is asserted.

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1 Results, sources, and conventions

The starting point is a collection of proved identities, not a blank research programme. The quartic reconstruction report [5] already evaluates all regular centered quartic trigamma moments. The later Bernoulli-cancellation report [6] proves the all-power recurrence, gives fifth- and sixth-power generators, and extends the mechanism to repeated even-order tails. The lattice-contact report [7] further treats arbitrary unequal even-order products. Those theorems must not be presented as new here.

The present continuation makes three different contributions. First, it reduces the named higher anchors J_7, K_8, K_9 explicitly. Second, it gives a cutoff-normalized compiler for arbitrary integer tail orders, with a parity-refined weight ceiling and a particularly simple odd-order mixed family. Third, it connects the successive differences of these tail products to Stieltjes constants, Gamma ratios, and order derivatives of multiple zeta functions. The transport identities use classical summation by parts and shift recurrences; the contribution is their explicit combined realization and normalization, not the underlying summation principle.

The most elementary new specialization is worth displaying first. Put

$$A_n = \zeta(2, n+1) = \psi'(n+1), \qquad B_n = \zeta(3, n+1) = -\frac{1}{2}\psi^{(2)}(n+1), \qquad x_n = n + \frac{1}{2}. \tag{1.1}$$

Then

$$\boxed{\sum_{n=0}^{\infty} \Big[(23x_n + 232x_n^3 + 240x_n^5)A_nB_n - 120x_n^2 - 76\Big] = -15.} \tag{1.2}$$

The bracket is a single absolutely summable quantity. None of its polynomial pieces is summed separately. Section 5 proves this by an explicit rational telescoper, independently of the multiple-zeta reduction software.

1.1 A scoped source audit

The core anchor definitions were read at the immutable repository ref `0dfef47533f8`; the supplementary unequal-even-tail report was inspected at `cc569dca8622`. Full immutable identifiers are recorded in the provenance file and the pinned bibliography links. The earlier quartic source and the canonical status files were also read. Exact paths and the scope of each read are in `SOURCE_AUDIT.md` and `provenance.json`.

The incoming directory listing and its intake instructions were inspected, but its binary archives were not exhaustively extracted or audited in this session. In particular, absence from the inspected passages is not evidence of absence from every incoming report. The article therefore offers explicit proved continuations and integration candidates, not a claim that every surrounding open question has been surveyed or resolved.

Classical ingredients are credited: Hurwitz asymptotics and Gamma relations [1], products of zeta tails ordered by their minimum index [2], and double-shuffle relations [3]. The existing fifth- and sixth-power generators and their original formal-coordinate ranks retain their source attribution.

1.2 Notation

We use descending-index multiple zeta values

$$\zeta(s_1, \dots, s_d) = \sum_{n_1 > \cdots > n_d \geq 1} \frac{1}{n_1^{s_1} \cdots n_d^{s_d}}.$$

An integer word is *admissible* when $s_1 \geq 2$ and all other indices are positive. Its weight is $s_1+\cdots+s_d$. The empty word has zeta value 1. Products can always be expanded into a finite linear combination of MZVs by the harmonic, or stuffle, product. No independence of the resulting numerical values is assumed.

Bernoulli polynomials satisfy

$$\frac{te^{xt}}{e^t-1}=\sum_{j\geq 0}B_j(x)\frac{t^j}{j!},\qquad B_j=B_j(0),\qquad B_1=-\frac{1}{2}.$$

For generalized Stieltjes constants our convention is

$$\zeta(1+u,a)=\frac{1}{u}+\sum_{q\geq 0}\frac{(-1)^q}{q!}\gamma_q(a)u^q,\qquad \gamma_0(a)=-\psi(a). \tag{1.3}$$

Order derivatives of zeta functions are always distinguished from argument derivatives. Positive real shifts suffice for every Stieltjes and Gamma identity below. This avoids any implicit choice of logarithmic branches.

2 Explicit reduction of the higher moment anchors

For a composition $(a_1,\dots,a_\ell)$ of e, write $(a_1,\dots,a_\ell)\vDash e$. Define the convergent anchor

$$\mathcal{Z}_{e,h}=\sum_{(a_1,\dots,a_\ell)\vDash e}\frac{e!}{a_1!\cdots a_\ell!}\zeta(2a_1,\dots,2a_{\ell-1},2a_\ell-h). \tag{2.1}$$

Only the negative h instances occurring below are needed here. Ordering the equal and strictly ordered tail indices gives the ordinary identity

$$\mathcal{Z}_{e,h}=\sum_{n=1}^{\infty}n^h(A_{n-1}^e-A_n^e),\qquad h\leq 0. \tag{2.2}$$

This ordering argument is classical [2]; the explicit reductions that follow are the additional step.

The Bernoulli-cancellation source defines

$$J_7=\mathcal{Z}_{2,-3}=2\zeta(2,5)+\zeta(7), \tag{2.3}$$

$$K_8=\frac{5}{6}\mathcal{Z}_{3,-2}+\frac{289}{32}\zeta(8), \tag{2.4}$$

$$K_9=\mathcal{Z}_{2,-5}-5\mathcal{Z}_{4,-1}. \tag{2.5}$$

In particular,

$$\mathcal{Z}_{3,-2}=6\zeta(2,2,4)+3\zeta(2,6)+3\zeta(4,4)+\zeta(8).$$

These definitions are inherited, rather than newly introduced unknown constants.

Theorem 2.1 (Three explicit anchor reductions)**.** *The anchors in* (2.3)–(2.5) *satisfy*

$$J_7=21\zeta(7)-8\zeta(2)\zeta(5)-4\zeta(3)\zeta(4), \tag{2.6}$$

$$K_8=10\zeta(2)\zeta(3)^2-50\zeta(3)\zeta(5)-10\zeta(5,3)+\frac{4927}{96}\zeta(8), \tag{2.7}$$

$$K_9=1086\zeta(9)-222\zeta(2)\zeta(7)+171\zeta(3)\zeta(6)-858\zeta(4)\zeta(5). \tag{2.8}$$

These are identities of convergent real series. They do not assert that $\zeta(5,3)$ is irreducible in any larger class of constants.

Proof. Expand the left sides by (2.1), and expand products on the right by the stuffle product. At each weight the difference is a rational vector on the finite set of admissible words. Appendix A specifies a finite family of true double-shuffle and duality relations and gives every coefficient expressing these three difference vectors as sums of those relations. The certificates use respectively 3, 17, and 54 nonzero rows. Their rational vector identities are checked independently by `verification/exact.py`; the complete row convention and coefficient tables are also printed in the appendix.

Each row is a genuine MZV identity, and the stated rational linear combination is exactly the target difference, so the difference vanishes. This proof uses neither integer-relation recognition nor a conjectural completeness theorem for double shuffle. □

For comparison, the exact computation gives the convenient equivalent anchor values

$$\mathcal{Z}_{3,-2} = 12\zeta(2)\zeta(3)^2 - 60\zeta(3)\zeta(5) - 12\zeta(5,3) + \frac{203}{4}\zeta(8), \tag{2.9}$$

$$\mathcal{Z}_{4,-1} = 42\zeta(2)\zeta(7) - 35\zeta(3)\zeta(6) + 170\zeta(4)\zeta(5) - 210\zeta(9). \tag{2.10}$$

The first follows immediately from (2.4) and (2.7). For the second, the classical odd-weight double-zeta reduction, or the same finite row system, gives

$$\mathcal{Z}_{2,-5} = 36\zeta(9) - 12\zeta(2)\zeta(7) - 4\zeta(3)\zeta(6) - 8\zeta(4)\zeta(5).$$

Insert this into (2.5). The auxiliary reduction can also be reconstructed with the included exact reducer; it is not inferred from a decimal fit.

2.1 Consequences for the existing quintic and sextic generators

Retain the inherited elementary combinations

$$Q_5 = 4\zeta(2)\zeta(3) - 10\zeta(5), \tag{2.11}$$

$$Q_7 = 51\zeta(3)\zeta(4) + 10\zeta(2)\zeta(5) - \frac{301}{4}\zeta(7). \tag{2.12}$$

The source proves $\sum_{n\geq 0} A_n^4 = Q_7$ and

$$\sum_{n\geq 0} x_n A_n^5 = -K_8 + \frac{5}{2}Q_7. \tag{2.13}$$

Thus the additional arithmetic reduction yields

$$\begin{aligned}\sum_{n\geq 0} x_n A_n^5 = {} & \frac{255}{2}\zeta(3)\zeta(4) + 25\zeta(2)\zeta(5) - \frac{1505}{8}\zeta(7) \\ & - 10\zeta(2)\zeta(3)^2 + 50\zeta(3)\zeta(5) + 10\zeta(5,3) - \frac{4927}{96}\zeta(8).\end{aligned} \tag{2.14}$$

The inherited moment relation (2.13) is explicitly credited; its expansion into the standard coordinates in (2.14) is the continuation.

For clarity, let I^q mean the q-fold primitive anchored at zero, and let $H_d = 2\sinh(t/2)G_d$ be the source's regular moment generator. Its sextic formula is

$$\begin{aligned}H_6(t) = {} & R_{11}t + 6I^2(\cosh(t/2)G_5(t)) - \frac{5}{2}I^4H_4(t) + \frac{1}{2}I^8H_2(t) \\ & + \frac{K_9}{12}t^3 + \frac{J_7}{240}t^5 - \frac{Q_5}{10080}t^7, \qquad R_{11} = \mathcal{Z}_{6,1}.\end{aligned} \tag{2.15}$$

Equations (2.6) and (2.8) eliminate the two named lower anchors here. The weight-eleven anchor R_{11} is not reduced in this article. Nor does rewriting the coordinates prove independence of the source's eleven numerical values.

3 Arbitrary integer-order tail products and regular parity

Fix a nonempty tuple $\boldsymbol{p} = (p_1, \dots, p_d)$ of integers $p_i \geq 2$. Define

$$T_n = \prod_{i=1}^{d} \zeta(p_i, n+1), \qquad W = \sum_{i=1}^{d} p_i, \qquad R = W - d, \qquad x_n = n + \frac{1}{2}. \tag{3.1}$$

Here $R \geq 1$. Odd orders are allowed. For integer $p \geq 2$,

$$\zeta(p, n+1) = \frac{(-1)^p}{(p-1)!} \psi^{(p-1)}(n+1),$$

so every result is simultaneously an identity for products of polygamma functions.

Set

$$\eta_{p,0} = \frac{1}{p-1}, \qquad \eta_{p,j} = \frac{(p)_{2j-1}}{(2j)!} B_{2j}(1/2) \quad (j \geq 1), \qquad c_k = \sum_{k_1+\cdots+k_d=k} \prod_i \eta_{p_i,k_i}. \tag{3.2}$$

The centered Hurwitz expansion [1] is

$$T_n \sim \sum_{k \geq 0} c_k x_n^{-R-2k}. \tag{3.3}$$

It is an asymptotic expansion, not a convergent series at a fixed n.

Proposition 3.1 (Poles and ordinary regular moments)**.** *The series*

$$\mathcal{D}_{\boldsymbol{p}}(s) = \sum_{n \geq 0} T_n x_n^{-s}$$

converges absolutely for $\Re s > 1 - R$ and continues meromorphically to the plane. Its poles are exactly the simple poles $s = 1 - R - 2k$, with residue $c_k \neq 0$. For each nonnegative integer $m \equiv R \pmod 2$, the regular value $M_{\boldsymbol{p},m} = \mathcal{D}_{\boldsymbol{p}}(-m)$ is the ordinary absolutely convergent sum

$$\boxed{M_{\boldsymbol{p},m} = \sum_{n=0}^{\infty} \left[x_n^m T_n - \sum_{0 \leq k \leq (m-R)/2} c_k x_n^{m-R-2k} \right].} \tag{3.4}$$

A sum whose upper bound is negative is empty.

Proof. A finite truncation of (3.3), with its remainder estimate, writes $\mathcal{D}$ as a normally convergent remainder series plus finitely many functions $c_k \zeta(s+R+2k, 1/2)$. Increasing the truncation continues it to successively larger left half-planes. The pole and residue assertions follow. Since $\eta_{p,j}$ has sign $(-1)^j$, every term in the convolution for c_k has sign $(-1)^k$; hence c_k is nonzero.

At $s = -m$ in the indicated parity, every removed term has added Hurwitz value $\zeta(-2j, 1/2) = 0$. The remainder in the bracket is $O(x_n^{-2})$, or faster when no subtraction is needed. This proves (3.4), with no unspecified summation of divergent polynomial series. □

Theorem 3.2 (Finite mixed-order reduction and refined weight ceiling)**.** *For every tuple in (3.1) and every regular degree m, there is a terminating rational algorithm expressing $M_{\boldsymbol{p},m}$ as a linear combination of 1 and admissible MZVs of weight at most $W-1$. If m is odd, the bound improves to $W-2$. The coefficient algorithm in Section 4 specifies the value without choosing a conjectural basis of MZVs.*

The all-even case overlaps the source's Bernoulli recurrences, which already give finite-span conclusions. The theorem's useful additional features are its allowance of arbitrary odd orders, its explicit cutoff-normalized finite procedure, and the uniform odd-degree refinement. The weight bound is a spanning statement, not a claim that a given value has that minimal weight or depth.

4 A finite-cutoff proof and an executable compiler

4.1 Harmonic words and their asymptotic coefficients

For a word $\boldsymbol{w} = (w_1, \dots, w_\ell)$ put

$$h_{\boldsymbol{w}}(N) = \sum_{N > n_1 > \cdots > n_\ell \geq 1} n_1^{-w_1} \cdots n_\ell^{-w_\ell}, \qquad h_\varnothing(N) = 1.$$

This is an *exclusive* cutoff. Also write $H_n(\boldsymbol{w}) = h_{\boldsymbol{w}}(n+1)$. First suppose every $w_i \geq 2$. Then

$$h_{\boldsymbol{w}}(N) \sim \sum_{h \geq 0} a_{\boldsymbol{w},h} N^{-h}, \qquad a_{\boldsymbol{w},0} = \zeta(\boldsymbol{w}). \tag{4.1}$$

There are no logarithmic terms in this expansion.

Let $z_{q,h}$ denote the coefficient of N^{-h} in $\zeta(q, N)$ for $q \geq 2$. Explicitly,

$$z_{q,h} = \begin{cases} (q-1)^{-1}, & h = q-1, \\ 1/2, & h = q, \\ B_{2j}(q)_{2j-1}/(2j)!, & h = q+2j-1,\ j \geq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{4.2}$$

The initial conditions and recurrence

$$a_{\varnothing,0} = 1, \quad a_{\varnothing,h} = 0\ (h > 0), \qquad a_{(s,\boldsymbol{v}),h} = -\sum_{r \geq 0} a_{\boldsymbol{v},r} z_{s+r,h} \quad (h > 0) \tag{4.3}$$

compute every required coefficient by finite arithmetic. For fixed h only finitely many r contribute.

Lemma 4.1. *Equations (4.1)–(4.3) are valid as asymptotic expansions with remainders to every fixed order. For $h > 0$, $a_{\boldsymbol{w},h}$ is a rational linear combination of zeta values of proper suffixes of $\boldsymbol{w}$.*

Proof. For a nonempty word,

$$\zeta(s, \boldsymbol{v}) - h_{(s,\boldsymbol{v})}(N) = \sum_{n \geq N} n^{-s} h_{\boldsymbol{v}}(n).$$

Insert any finite asymptotic expansion for the shorter word. Since $s \geq 2$, its remainder can be summed absolutely and gives an error of the next required order. The retained terms are $a_{\boldsymbol{v},r}\zeta(s+r, N)$. Expanding each of them gives (4.3). Induction on the word length proves all assertions, including the suffix statement. □

4.2 Cutoff constants with a nonpositive or unit first index

The finite summation step will introduce a possibly nonpositive first index, but all following indices remain at least two. Let $\mathcal{C}(\boldsymbol{w})$ denote the coefficient of $N^0(\log N)^0$ in the expansion of $h_{\boldsymbol{w}}(N)$. For an admissible word it is simply $\zeta(\boldsymbol{w})$. The following finite formulas define all other cases that are needed.

For a suffix $\boldsymbol{v} = (v_1, \dots, v_\ell)$ with $v_i \geq 2$,

$$\begin{aligned}\mathcal{C}(1, \boldsymbol{v}) = \gamma\zeta(\boldsymbol{v}) &- \sum_{j=1}^{\ell} \zeta(v_1, \dots, v_j, 1, v_{j+1}, \dots, v_\ell) \\ &- \sum_{j=1}^{\ell} \zeta(v_1, \dots, v_j + 1, \dots, v_\ell).\end{aligned} \tag{4.4}$$

For the empty suffix this means $\mathcal{C}(1) = \gamma$. All the other displayed MZVs converge.

For $r \geq 0$ define the Faulhaber polynomial

$$P_r(X) = \sum_{k=1}^{X} k^r = \frac{B_{r+1}(X+1) - B_{r+1}(1)}{r+1} = \sum_{\ell \geq 1} p_{r,\ell} X^\ell. \tag{4.5}$$

The expression is polynomial notation, not a restriction that X be an integer. If the suffix is empty, set

$$\mathcal{C}(-r) = P_r(-1) = -\mathbf{1}_{r=0}. \tag{4.6}$$

If the suffix is $(s, \boldsymbol{u})$, set

$$\mathcal{C}(-r, s, \boldsymbol{u}) = \mathrm{CT}_N\Big[P_r(N-1)h_{(s,\boldsymbol{u})}(N)\Big] - \sum_{\ell \geq 1} p_{r,\ell}\mathcal{C}(s - \ell, \boldsymbol{u}). \tag{4.7}$$

The constant term of a polynomial times h is obtained directly from the coefficients in (4.3). Each recursive call in (4.7) has shorter length.

Lemma 4.2 (Exact meaning of the cutoff constants)**.** *Equations* (4.4)–(4.7) *equal the actual asymptotic cutoff constants. Their recursion terminates and introduces only rational numbers, admissible MZVs, and a term linear in* γ*.*

Proof. The finite stuffle identity for $h_1(N)h_{\boldsymbol{v}}(N)$ isolates $h_{1,\boldsymbol{v}}(N)$. All other terms converge; $h_1(N) = \log N + \gamma + O(N^{-1})$ and $h_{\boldsymbol{v}}(N) = \zeta(\boldsymbol{v}) + O(N^{-1})$. Taking the constant term gives (4.4).

For a nonpositive first index, reverse the finite sums:

$$h_{(-r,s,\boldsymbol{u})}(N) = P_r(N-1)h_{(s,\boldsymbol{u})}(N) - \sum_{\ell \geq 1} p_{r,\ell} h_{(s-\ell,\boldsymbol{u})}(N).$$

This is an exact identity for every positive integer N. Taking cutoff constants gives (4.7). The empty suffix gives $P_r(N-1)$ itself, whose constant is $P_r(-1)$. Induction proves termination and the stated form of the answer. □

4.3 The master formula

Multiplying the finite identities $\zeta(p, n+1) = \zeta(p) - H_n^{(p)}$ gives

$$T_n = \sum_{\boldsymbol{w}} b_{\boldsymbol{w}} H_n(\boldsymbol{w}). \tag{4.8}$$

All entries in every nonempty $\boldsymbol{w}$ are at least two. The coefficients $b_{\boldsymbol{w}}$ are finite polynomials in ordinary zeta values. They are generated by the stuffle product, starting from $b_\varnothing = 1$ and multiplying one factor $\zeta(p) - H_n^{(p)}$ at a time. Every homogeneous term satisfies $\mathrm{wt}(b_{\boldsymbol{w}}) + \mathrm{wt}(\boldsymbol{w}) = W$.

Define the centered discrete primitive

$$F_m(X) = \frac{B_{m+1}(X + 1/2) - B_{m+1}(1/2)}{m+1} = \sum_{h \geq 1} f_{m,h} X^h. \tag{4.9}$$

It has zero constant term and $F_m(n+1) - F_m(n) = x_n^m$.

Theorem 4.3 (Finite coefficient formula)**.** *For the regular degrees in Proposition 3.1,*

$$\boxed{M_{\boldsymbol{p},m} = \sum_{\boldsymbol{w} \neq \varnothing} b_{\boldsymbol{w}} \sum_{h \geq 1} f_{m,h} \left[a_{\boldsymbol{w},h} - \mathcal{C}(w_1 - h, w_2, \ldots, w_\ell) \right].} \tag{4.10}$$

All sums are finite. After the stuffle product is expanded, the coefficient of γ is zero.

Proof. For a nonempty word, reversing the finite sums gives

$$\sum_{n=0}^{N-1} x_n^m H_n(\boldsymbol{w}) = F_m(N) h_{\boldsymbol{w}}(N) - \sum_{h \geq 1} f_{m,h} h_{(w_1 - h, w_2, \ldots, w_\ell)}(N). \tag{4.11}$$

Indeed, the contribution with largest harmonic index k has the inner sum $\sum_{n=k}^{N-1} x_n^m = F_m(N) - F_m(k)$. Insert (4.8) and take cutoff constants. The empty word contributes $F_m(N)$, whose constant is zero. Lemmas 4.1 and 4.2 give precisely (4.10).

It remains to identify this constant with the ordinary sum. Every subtracted monomial in (3.4) is an even power of x_n. Its finite sum is $F_{2j}(N)$, whose constant is zero. The sum of the remaining bracket tends to the ordinary value, with an $O(N^{-1})$ remainder. Thus the original cutoff constant is that value.

Finally, regular parity makes every exponent in the centered asymptotic summand even. Its partial sums have no logarithmic term. In the finite reduction, every occurrence of γ comes from $h_1(N) = \log N + \gamma + o(1)$ and carries exactly the same coefficient as the corresponding logarithm. That total coefficient is zero. This proves the cancellation without treating γ as an independent transcendental number. □

Proof of Theorem 3.2. The recursions decrease either the suffix length or a bounded coefficient index, so they terminate. For $h > 0$, the term $a_{\boldsymbol{w},h}$ contains only proper suffixes and hence has weight at most $\mathrm{wt}(\boldsymbol{w}) - 2$. If $w_1 - h \geq 1$, the non-γ terms in the corresponding cutoff constant have weight $\mathrm{wt}(\boldsymbol{w}) - h$. If $w_1 - h \leq 0$, the negative-index recursion gives weights at most the weight of the remaining suffix, again no more than $\mathrm{wt}(\boldsymbol{w}) - 2$. This last bound follows inductively from (4.7); its polynomial boundary term uses only suffix values, and each subsequent positive leading index is reduced rather than increased.

Multiplying by $b_{\boldsymbol{w}}$ and expanding products preserves total weight. Since $h \geq 1$, this gives the bound $W - 1$. When m is odd, F_m is an even polynomial with zero constant term, so every nonzero $f_{m,h}$ has $h \geq 2$. The same argument then gives $W - 2$. □

5 The complete odd-degree (2,3) family

The general compiler simplifies sharply for the product in (1.1). Its total weight is 5, so all its regular moments lie in $\mathrm{span}_{\mathbb{Q}}\{1, \zeta(2), \zeta(3)\}$. A direct formula computes these three coefficients without a general MZV reduction.

Let $P(X) = \sum_{j\geq 1} p_{2j} X^{2j}$ be any even rational polynomial with $P(0) = 0$, and set

$$W_P(x) = P(x + 1/2) - P(x - 1/2), \qquad \mathscr{M}(P) = \sum_{m \text{ odd}} [x^m] W_P(x)\, M_{(2,3),m}.$$

The last expression means one ordinary bracketed sum with the subtractions prescribed in (3.4). Put

$$\lambda(P) = \sum_{j\geq 1} p_{2j} B_{2j-2}, \qquad Q(X) = P(X) - p_2 X^2, \tag{5.1}$$

and define rational polynomials by finite power sums:

$$R_Q(N) = \sum_{k=1}^{N} \frac{Q(k)}{k^3}, \qquad S_Q(N) = \sum_{k=1}^{N} \frac{Q(k)}{k^2}. \tag{5.2}$$

They are polynomials because Q is divisible by X^4.

Theorem 5.1 (All odd-degree mixed coefficients)**.** *With the preceding definitions,*

$$\boxed{\mathscr{M}(P) = p_2\zeta(3) + \lambda(P)\zeta(2) + \rho(Q),} \tag{5.3}$$

where the rational number $\rho(Q)$ is given by the finite coefficient extraction

$$\rho(Q) = \mathrm{CT}_N \left[Q(N) A_N B_N + R_Q(N) A_N + S_Q(N) B_N \right]. \tag{5.4}$$

The asymptotic expansions of A_N and B_N at infinity make (5.4) a finite Bernoulli calculation. In particular, $p_2 = \lambda(P) = 0$ gives a rational ordinary sum.

Proof. First assume $p_2 = 0$ and write $P = Q$. Define $V_n = P(n)A_nB_n + R_P(n)A_n + S_P(n)B_n$. The shifts $A_{n+1} = A_n - (n+1)^{-2}$ and $B_{n+1} = B_n - (n+1)^{-3}$ give

$$V_{n+1} - V_n = W_P(x_n) A_n B_n + E_P(n+1), \tag{5.5}$$

where

$$E_P(y) = \frac{P(y)}{y^5} - \frac{R_P(y)}{y^2} - \frac{S_P(y)}{y^3}.$$

Faulhaber's formula shows that this Laurent polynomial has no y^{-1} term and that its y^{-2} coefficient is $-\lambda(P)$. For completeness, the linear coefficient of R_P is $p_4/2$, the quadratic coefficient of S_P is $p_4/2$, and the linear coefficient of S_P is $\lambda(P)$. The possible y^{-1} coefficient is therefore $p_4 - p_4/2 - p_4/2 = 0$. Thus $E_P(y) = E_P^{\mathrm{pol}}(y) - \lambda(P) y^{-2}$.

Sum (5.5) for $0 \leq n < N$ and take cutoff constants. The finite sum of the polynomial $E_P^{\mathrm{pol}}(k)$, $1 \leq k \leq N$, is a polynomial in N with zero constant. Also $V_0 = 0$ and $H_N^{(2)} \to \zeta(2)$. The ordinary-sum normalization from Proposition 3.1 now gives $\mathscr{M}(P) = \rho(P) + \lambda(P)\zeta(2)$.

For $P(X) = X^2$, finite summation by parts reduces the value to

$$\sum_{k\geq 1} \frac{A_k}{k} + \sum_{k\geq 1} B_k + \zeta(3) = \zeta(2,1) + \zeta(2) - \zeta(3) + \zeta(3) = \zeta(2) + \zeta(3).$$

We used the classical Euler identity $\zeta(2,1) = \zeta(3)$. Adding p_2 times this case proves (5.3). □

Taking $P = F_1, F_3, F_5$ gives, respectively,

$$\sum_{n\geq 0} x_n A_n B_n = \frac{\zeta(2)+\zeta(3)}{2}, \tag{5.6}$$

$$\sum_{n\geq 0}\left(x_n^3 A_n B_n - \frac{1}{2}\right) = -\frac{\zeta(2)}{12} - \frac{\zeta(3)}{8} - \frac{5}{48}, \tag{5.7}$$

$$\sum_{n\geq 0}\left(x_n^5 A_n B_n - \frac{x_n^2}{2} + \frac{1}{6}\right) = \frac{47\zeta(2)}{1440} + \frac{7\zeta(3)}{96} + \frac{11}{288}. \tag{5.8}$$

Theorem 5.1 is an all-degree result; these examples are not the basis of an extrapolation.

5.1 A direct finite proof of the rational identity

The combination $23F_1 + 232F_3 + 240F_5$ simplifies to $P(X) = 40X^6 + 8X^4$. Its p_2 is zero and $\lambda(P) = 40B_4 + 8B_2 = 0$. More explicitly, define

$$\begin{aligned} U_n = {} & 8n^4(5n^2+1)A_nB_n + 2n(n+1)(5n^2+5n+2)A_n \\ & + 4n^2(n+1)^2(2n+1)B_n - 34n^3 + 9n^2 - 33n. \end{aligned} \tag{5.9}$$

Direct substitution of the two tail shifts gives the exact identity

$$U_{n+1} - U_n = (23x_n + 232x_n^3 + 240x_n^5)A_nB_n - 120x_n^2 - 76. \tag{5.10}$$

Here $U_0 = 0$. The ordinary uncentered expansions are

$$\begin{aligned} A_n &= n^{-1} - \tfrac{1}{2}n^{-2} + \tfrac{1}{6}n^{-3} - \tfrac{1}{30}n^{-5} + O(n^{-7}), \\ B_n &= \tfrac{1}{2}n^{-2} - \tfrac{1}{2}n^{-3} + \tfrac{1}{4}n^{-4} - \tfrac{1}{12}n^{-6} + O(n^{-8}). \end{aligned}$$

They give $U_n = -15 + 11/(3n) + O(n^{-2})$. Summing (5.10) proves (1.2). Absolute convergence also follows directly from the centered subtraction, whose remainder is $O(x_n^{-2})$.

Equivalently, in polygamma notation,

$$\begin{aligned} \sum_{n\geq 0}\big[& (23x_n + 232x_n^3 + 240x_n^5)\psi'(n+1)\psi^{(2)}(n+1) \\ & + 240x_n^2 + 152\big] = 30. \end{aligned} \tag{5.11}$$

The sign here follows from $B_n = -\psi^{(2)}(n+1)/2$.

5.2 Some further genuinely mixed examples

The same finite compiler gives the following exact identities. They illustrate odd-order participation and are accompanied by full coefficient records in `results/mixed_moments.json`:

$$\sum_{n\geq 0} A_n^2 B_n = \frac{3}{2}\zeta(3)^2 + \frac{7}{6}\zeta(6), \tag{5.12}$$

$$\sum_{n\geq 0} x_n^2 A_n^2 B_n = \frac{1}{3}\zeta(2) + \frac{4}{3}\zeta(3) - \frac{7}{12}\zeta(4) - \frac{1}{8}\zeta(3)^2 - \frac{7}{72}\zeta(6), \tag{5.13}$$

$$\sum_{n\geq 0}\left(x_n^4 A_n^2 B_n - \frac{1}{2}\right) = -\frac{3}{20}\zeta(2) - \frac{7}{15}\zeta(3) + \frac{7}{24}\zeta(4) + \frac{7}{160}\zeta(3)^2 + \frac{49}{1440}\zeta(6) - \frac{3}{40}. \tag{5.14}$$

With $C_n = \zeta(4, n+1) = \psi^{(3)}(n+1)/6$, an unweighted weight-eight example is

$$\sum_{n\geq 0} A_n B_n C_n = -\frac{15}{2}\zeta(2)\zeta(3)^2 + 30\zeta(3)\zeta(5) + \frac{9}{2}\zeta(5,3) - \frac{1261}{72}\zeta(8). \tag{5.15}$$

All unsubtracted sums displayed here converge absolutely. The proofs are Theorem 4.3 followed by the finite classical relation system of Appendix A; the exact verifier regenerates these outputs. The unweighted index-ordering principle itself is not new [2].

6 A finite polylogarithm kernel and final-index zeta derivatives

Return to an arbitrary tuple $\boldsymbol{p}$. For $|z| < 1$ set

$$\mathrm{Li}_{w_1,\dots,w_\ell}(z) = \sum_{n_1>\cdots>n_\ell\geq 1} \frac{z^{n_1}}{n_1^{w_1}\cdots n_\ell^{w_\ell}}, \qquad \mathrm{Li}_\varnothing(z) = 1,$$

and use the coefficients from (4.8) to define the finite kernel

$$\mathcal{F}_{\boldsymbol{p}}(z) = \sum_{\boldsymbol{w}} b_{\boldsymbol{w}} \,\mathrm{Li}_{\boldsymbol{w}}(z). \tag{6.1}$$

Its depth is at most d in this explicit representation; no minimal functional depth is claimed.

Proposition 6.1 (The ordinary generating function)**.** *For $|z| < 1$,*

$$\boxed{\sum_{n\geq 0} T_n z^n = \frac{\mathcal{F}_{\boldsymbol{p}}(z)}{1-z}.} \tag{6.2}$$

For example,

$$\begin{aligned}\mathcal{F}_{2,3}(z) = {}& \zeta(2)\zeta(3) - \zeta(3)\,\mathrm{Li}_2(z) - \zeta(2)\,\mathrm{Li}_3(z) \\ & + \mathrm{Li}_{2,3}(z) + \mathrm{Li}_{3,2}(z) + \mathrm{Li}_5(z).\end{aligned} \tag{6.3}$$

Proof. The absolutely convergent identity $\sum_{n\geq 0} H_n(\boldsymbol{w}) z^n = \mathrm{Li}_{\boldsymbol{w}}(z)/(1-z)$ follows by summing first over $n \geq n_1$. Apply it to the finite expansion (4.8). Formula (6.3) uses $H_n^{(2)} H_n^{(3)} = H_n(2,3) + H_n(3,2) + H_n^{(5)}$. □

Define a holomorphic Dirichlet family on $\Re s > 1 - R$ by

$$\mathcal{E}_{\boldsymbol{p}}(s) = \sum_{n\geq 1} \frac{T_{n-1}}{n^s}. \tag{6.4}$$

For an ordered set partition $\pi = (B_1, \dots, B_\ell)$ of $\{1, \dots, d\}$, let $w_{\pi,j} = \sum_{i\in B_j} p_i$. Every ordered partition is included once, even when different partitions give the same word.

Theorem 6.2 (MZV and Mellin realizations)**.** *On $\Re s > 1 - R$,*

$$\mathcal{E}_{\boldsymbol{p}}(s) = \sum_{\pi} \left[\zeta(w_{\pi,1}, \dots, w_{\pi,\ell}, s) + \zeta(w_{\pi,1}, \dots, w_{\pi,\ell-1}, w_{\pi,\ell} + s)\right]. \tag{6.5}$$

Every series on the right converges there. On $\Re s > 0$ it also has the ordinary integral representation

$$\boxed{\mathcal{E}_{\boldsymbol{p}}(s) = \frac{1}{\Gamma(s)} \int_0^1 \frac{\mathcal{F}_{\boldsymbol{p}}(z)}{1-z} (-\log z)^{s-1}\, dz.} \tag{6.6}$$

The identities may be differentiated any finite number of times in s on compact subsets of their respective domains.

Proof. In T_{n-1} every tail index is at least n. Group equal tail indices and order their blocks decreasingly. The distinguished index n is either strictly below the last block or equal to it. These are exactly the two terms in (6.5). Absolute convergence follows from $T_{n-1} = O(n^{-R})$; the same bounds with fixed powers of $\log n$ give local uniformity of all derivatives. The individual multiple series satisfy the same convergence conditions: all proper initial block sums are at least twice the number of blocks, and the full sum satisfies the condition because $\Re s > 1 - R$.

For the integral, use $n^{-s} = \Gamma(s)^{-1} \int_0^1 z^{n-1}(-\log z)^{s-1}\, dz$ and (6.2). Absolute convergence justifies the exchange. Near $z = 1$, the generating function is bounded when $R > 1$ and is at most logarithmic when $R = 1$. Near zero it is bounded. These endpoint bounds also justify all order derivatives for $\Re s > 0$. □

At $s = 1$ a particularly useful all-index form uses the polynomials

$$\mathscr{P}_q(y) = q![u^q]\frac{e^{uy}}{\Gamma(1+u)}. \tag{6.7}$$

For example,

$$\mathscr{P}_0 = 1, \quad \mathscr{P}_1 = y + \gamma, \quad \mathscr{P}_2 = (y+\gamma)^2 - \zeta(2),$$
$$\mathscr{P}_3 = (y+\gamma)^3 - 3\zeta(2)(y+\gamma) + 2\zeta(3).$$

Differentiating (6.6) gives

$$\mathcal{E}_{\boldsymbol{p}}^{(q)}(1) = \int_0^1 \frac{\mathcal{F}_{\boldsymbol{p}}(z)}{1-z}\mathscr{P}_q(\log(-\log z))\, dz. \tag{6.8}$$

All cancellations in $\mathcal{F}$ are kept inside the integral. The reciprocal-Gamma coefficient construction is classical; its use here fixes the constants in the tail-weighted Stieltjes identities.

7 Stieltjes sums, polygamma derivatives, and anchored primitives

Let

$$\Delta_n = T_{n-1} - T_n \quad (n \geq 1), \qquad Z_{\boldsymbol{p}} = T_0 = \prod_i \zeta(p_i). \tag{7.1}$$

These positive weights satisfy

$$\sum_{n\geq 1} \Delta_n = Z_{\boldsymbol{p}}, \qquad \sum_{n>k} \Delta_n = T_k, \qquad \Delta_n = O(n^{-R-1}). \tag{7.2}$$

Only these elementary tail properties are required for the first transport theorem. They follow either from telescoping or from $\zeta(p,n) - \zeta(p,n+1) = n^{-p}$.

Theorem 7.1 (Every Stieltjes index and argument)**.** *For every $q \geq 0$ and $a > 0$,*

$$\boxed{\sum_{n\geq 1} \Delta_n \gamma_q(n+a) = Z_{\boldsymbol{p}}\gamma_q(a) - \sum_{k\geq 0} T_k \frac{\log^q(k+a)}{k+a}.} \tag{7.3}$$

Both sums converge absolutely, locally uniformly for $a > 0$. At $a = 1$ this becomes the convergent final-index derivative identity

$$\sum_{n\geq 1} \Delta_n \gamma_q(n+1) = Z_{\boldsymbol{p}}\gamma_q(1) - (-1)^q \mathcal{E}_{\boldsymbol{p}}^{(q)}(1). \tag{7.4}$$

The finite MZV expression (6.5) and the polylogarithm integral (6.8) specify the derivative on the right.

Proof. Comparing Laurent coefficients in the Hurwitz shift equation gives

$$\gamma_q(n+a) = \gamma_q(a) - \sum_{k=0}^{n-1} \frac{\log^q(k+a)}{k+a}. \tag{7.5}$$

Multiply by Δ_n, sum, and use (7.2). The double sum of absolute values is bounded by $\sum_k T_k |\log^q(k+a)|/(k+a)$, which converges because $R \geq 1$. Also $\gamma_q(n+a) = O(1+\log^{q+1}(n+1))$, either directly from (7.5) or from Hurwitz asymptotics. Hence the left sum is absolutely convergent. The same estimates are uniform for a in any compact subinterval of $(0,\infty)$. Finally, differentiation of (6.4) at $s=1$ gives (7.4). □

For an elementary check, take one tail of order two, so $\Delta_n = n^{-2}$. The $q=0$ identity reduces to the classical Euler sum $\sum H_n/n^2 = 2\zeta(3)$. At $q=1$ it gives the ordinary derivative identity

$$\sum_{n\geq 1} \frac{\gamma_1(n+1)}{n^2} = \zeta(2)\gamma_1(1) + \frac{\partial}{\partial s}\zeta(2,s)\Big|_{s=1} + \zeta'(3). \tag{7.6}$$

No singular multiple-zeta regularization is needed: the differentiated series converges in a neighborhood of $s=1$.

Corollary 7.2 (Every polygamma order)**.** *For $r \geq 0$ and $a > 0$,*

$$\sum_{n\geq 1} \Delta_n \psi^{(r)}(n+a) = Z_{\boldsymbol{p}}\psi^{(r)}(a) + (-1)^r r! \sum_{k\geq 0} \frac{T_k}{(k+a)^{r+1}}. \tag{7.7}$$

Proof. For $r=0$ this is (7.3) with $q=0$ and $\gamma_0 = -\psi$. For general r, use the differentiated digamma shift equation and the same absolutely convergent interchange. □

7.1 Gamma ratios and all Stieltjes primitives

Theorem 7.3 (Anchored Gamma and Hurwitz-zero primitives)**.** *For $a, b > 0$,*

$$\sum_{n\geq 1} \Delta_n \log \frac{\Gamma(n+b)}{\Gamma(n+a)} = Z_{\boldsymbol{p}} \log \frac{\Gamma(b)}{\Gamma(a)} + \sum_{k\geq 0} T_k \log \frac{k+b}{k+a}. \tag{7.8}$$

For $q \geq 0$ set

$$\mathcal{J}_q(a,b) = \partial_s^{q+1}\zeta(s,b)\Big|_{s=0} - \partial_s^{q+1}\zeta(s,a)\Big|_{s=0}.$$

Then

$$\begin{aligned} &\sum_{n\geq 1} \Delta_n \mathcal{J}_q(n+a, n+b) - Z_{\boldsymbol{p}}\mathcal{J}_q(a,b) \\ &\qquad = (-1)^q \sum_{k\geq 0} T_k \left[\log^{q+1}(k+b) - \log^{q+1}(k+a)\right]. \end{aligned} \tag{7.9}$$

All these series converge absolutely, and their anchors vanish when $a=b$. The $q=0$ case of (7.9) is (7.8).

Proof. The finite Gamma product gives

$$\log \frac{\Gamma(n+b)}{\Gamma(n+a)} = \log \frac{\Gamma(b)}{\Gamma(a)} + \sum_{k=0}^{n-1} \log \frac{k+b}{k+a}.$$

The inner difference is $O((k+1)^{-1})$, so the same summation argument as before proves (7.8).

To prove the all-index primitive identity, differentiate $\partial_a \zeta(s,a) = -s\zeta(s+1,a)$ at $s=0$, using (1.3). One obtains

$$\partial_a \, \partial_s^{q+1} \zeta(s,a)\Big|_{s=0} = -(q+1)(-1)^q \gamma_q(a). \tag{7.10}$$

Integrate (7.3) from a to b and multiply by $-(q+1)(-1)^q$. The elementary logarithmic primitive gives (7.9). Uniform convergence on the compact interval between a and b justifies the integration. Finally, $\zeta'(0,a) = \log\Gamma(a) - \frac{1}{2}\log(2\pi)$ gives the $q=0$ identification [1]. □

7.2 A local analytic Gamma generator with explicit coefficients

Setting $a=1$, $b=1+z$ in (7.8) gives, initially for real $|z|<1$ and then holomorphically in that disk,

$$\begin{aligned} \sum_{n\ge 1} \Delta_n \log\frac{\Gamma(n+1+z)}{\Gamma(n+1)} - Z_{\boldsymbol{p}} \log\Gamma(1+z) \\ = \sum_{m\ge 1} \frac{(-1)^{m+1}}{m} \mathcal{E}_{\boldsymbol{p}}(m) z^m. \end{aligned} \tag{7.11}$$

Here the branch is the analytic logarithm fixed to zero at $z=0$. The coefficients are the finite convergent MZV combinations (6.5). Absolute convergence on $|z| \le r < 1$ follows from the logarithm series and $T_k = O((k+1)^{-R})$. This gives an anchored antiderivative family rather than a formula with unspecified additive constants.

8 Proof audit, computational evidence, and integration

8.1 What the finite checks establish

The delivered code has two logically distinct parts. The exact certificate verifier parses only restricted rational/MZV arithmetic, expands stuffle products, independently forms binary shuffles by interleaving positions, and reconstructs every cited relation. It checks that each target difference equals the stated rational combination exactly. A changed coefficient is deliberately rejected for each of the three main anchor certificates.

The mixed-moment tests independently check finite centered summation identities at rational cutoffs, the tail-product stuffle expansion, Euler-constant cancellation, the refined weight ceilings, agreement with the repeated-tail expansion, and the separate one-tail Faulhaber formula for the $(2,3)$ family. The -15 telescoper is checked as an identity of rational functions in n, A, B, not merely on sampled special-function values. The exact ranks of the stated finite row matrices are also recomputed. Their ranks concern rational relations on formal word coordinates, not dimensions of numerical period spaces.

The final replay passes 369 exact assertions, including three deliberate corruption controls. Complete records are in `results/exact.json`. The shortened certificates retain exactly the required nonzero relation rows; every retained row is reconstructed from its defining words. The numerical suite contains 40 comparisons at 90 working decimal digits, including the three anchor sums, multiple truncations of $\zeta(5,3)$ and fifteen mixed moments, finite Stieltjes and Gamma transport, and a polylogarithm integral. Their largest recorded residual is below 10^{-68} in the delivered run.

These numerical comparisons are diagnostics, *not* interval certificates. Tail acceleration uses finite asymptotic expansions at two separate truncation choices. The analytic proofs supply convergence

and equality; decimal agreement supplies an independent check of signs, constants, and normalization. There is no Lean or other proof-assistant formalization in this package.

8.2 A real normalization hazard, not an attributed source error

The exclusive cutoff matters. One has

$$h_{(0)}(N) = N - 1, \qquad \mathrm{CT}_N\, h_{(0)}(N) = -1,$$

whereas the inclusive cutoff $\sum_{n=1}^{N} 1 = N$ has constant zero. Replacing (4.6) by the inclusive value would silently change the finite moment compiler. The separate check $M_{(2),1} = -1/4$ detects such mistakes. This warning is not an allegation that the inspected predecessor made that substitution.

No mathematical error in the specific predecessor formulas used here was found in this audit. A confirmed editorial issue in the inspected canonical README is the verbatim repetition of the paragraph beginning with the complete 1,032-input first-letter-X family and its rank 1,008/kernel dimension 24. The proposed correction is simply to remove the duplicate copy if it remains in the integration baseline. The more important attribution correction in this research process was to retain the newer report's ownership of the all-power recurrence rather than relabel that theorem as a new result.

8.3 Repository placement

A natural report destination is the continuation `mixed-tail-anchor-reductions` under the existing `stieltjes-harmonic-resonance` spine, with a cross-reference from the mixed-tail material of the lattice-contact report. The package contains a short integration fragment, a claim-status map, and source-audit notes. It does not modify the remote repository, apply a canonical-manuscript patch, or change the status of the Gaussian S_6, revised S_8, or general unequal-tail reconstruction problems. Canonical integration should preserve the attribution boundaries set out in Section 1.

9 Further research questions

1. Reduce the remaining weight-eleven sextic anchor. The explicit value $R_{11} = \mathcal{Z}_{6,1}$ remains an admissible finite MZV combination. A natural next task is to reduce it into chosen weight-eleven coordinates, with a compact exact certificate, and then rewrite the entire sextic generator in those coordinates. No numerical independence is required to perform this reduction.

2. Find a mixed-order analytic generator with all anchors fixed. The compiler proves a finite arithmetic span for arbitrary integer orders, including odd orders. The even-order source has stronger analytic generating-function recurrences. Construct an equally explicit recurrence for arbitrary mixed parity, determine its necessary boundary terms, and give its finite multiple-polylogarithm realization with all endpoint constants specified.

3. Classify rational polynomial combinations. For the $(2, 3)$ family the two conditions $p_2 = \lambda(P) = 0$ produce rational values, with a finite telescoping construction. Find analogous coefficient conditions for other mixed products. The first concrete candidates are $(2, 2, 3)$ and $(2, 3, 4)$, whose small formal coordinate spaces are visible in the supplied tables. This is a rational coefficient-space problem, not a problem of proving arithmetic independence.

4. Introduce rational and unequal argument shifts. Replace $\zeta(p_i, n+1)$ by $\zeta(p_i, n+a_i)$, first for positive rational a_i. The centered polynomial subtraction no longer has all its Hurwitz corrections equal to zero. Determine a finite colored-MZV or Lerch-coordinate compiler retaining every correction, and identify which parity-refined weight statements survive.

5. Differentiate the tail orders themselves. The Stieltjes transport already gives final-index derivatives of MZVs. Differentiating the individual orders p_i would introduce logarithmic decorations on all ordered blocks. Prove a joint continuation and differentiation theorem with explicit correction terms at any new resonant parameter values. Finite experimental agreement should not substitute for that local-uniform analysis.

6. Formalize the finite and analytic layers separately. The restricted rational certificate checker and the finite summation identity (4.11) are suitable first targets for formalization. The next layer is the asymptotic remainder theorem and its conversion of cutoff constants into ordinary subtracted sums. Keeping these layers separate would prevent a finite rank certificate from being mistaken for an analytic or transcendence proof.

A Complete finite certificates for the three anchors

This appendix makes the finite proof of Theorem 2.1 reconstructible without any numerical MZV database. All row indices below are one-based. For weight w, list the admissible compositions of w in the order generated by choosing the first part increasingly, then the next part increasingly, recursively. This gives 2^{w-2} columns, one for each formal MZV word.

For an index word $u = (u_1, \dots, u_r)$, its binary encoding is $0^{u_1-1}1\cdots 0^{u_r-1}1$. Decode any binary word ending in 1 by counting the zero runs plus one. Let $u * v$ be the positive stuffle product of index words, and let $u \,\mathrm{sh}\, v$ denote the sum obtained by shuffling their binary encodings and decoding. Here sh is written in the tables' generating rule as the ordinary shuffle operation; repeated words retain their multiplicities.

The ordered relation list is constructed as follows. For $k = 1, \dots, w-2$, take $u = (1)$ if $k = 1$, and otherwise take every admissible composition u of k in the preceding order. For each u, take every admissible composition v of $w-k$ in that order. Append the coefficient vector of

$$\mathcal{R}(u, v) = \zeta(u * v) - \zeta(u \,\mathrm{sh}\, v), \tag{A.1}$$

omitting a row only when its entire coefficient vector is zero. After these rows, append $\zeta(u) - \zeta(u^\vee)$ for every admissible word u of weight w, in the same order, omitting self-dual words. The dual $u^\vee$ is obtained by reversing the binary encoding and exchanging zero and one. Duplicate nonzero rows are not removed.

For convergent u, v, (A.1) is the equality between the series product and the iterated-integral product. For $u = (1)$, the same row is the one-leading-one regularized double-shuffle identity: the divergent leading word cancels and the surviving combination is admissible. This special case can be obtained by subtracting the common logarithmic term before taking the endpoint limit; it is also a direct instance of [3]. Duality follows from reversal and $t \mapsto 1-t$ in the iterated integrals. Thus every row used is a proved identity.

Weight	Columns	Rows	Rational row rank	Rows in anchor certificate
7	32	80	29	3
8	64	168	60	17
9	128	384	123	54

Let D_7, D_8, D_9 be the left side minus the right side of (2.6), (2.7), (2.8), respectively, after the anchor definitions and all stuffle products are expanded. The tables give the exact vector identities

$$D_w = \sum_j c_j \mathcal{R}_{w,j}. \tag{A.2}$$

All unlisted coefficients are zero. Ranks in the preceding table are only diagnostics of this finite relation system; equality in (A.2), not a completeness conjecture, proves the three anchor identities.

Weight 7 certificate

j	c_j	j	c_j	j	c_j
16	-10	24	3	32	-1

Weight 8 certificate

j	c_j	j	c_j	j	c_j
24	$\frac{85}{33}$	28	$\frac{85}{11}$	30	$\frac{265}{33}$
31	$-\frac{125}{11}$	32	$\frac{635}{132}$	40	$\frac{415}{132}$
44	$\frac{995}{132}$	46	$\frac{301}{132}$	47	$-\frac{68}{33}$
48	$-\frac{1347}{88}$	56	$-\frac{250}{33}$	60	$-\frac{665}{132}$
62	$-\frac{71}{12}$	63	$\frac{343}{66}$	64	$\frac{5239}{264}$
72	$\frac{895}{132}$	80	$-\frac{951}{88}$		

Weight 9 certificate

j	c_j	j	c_j	j	c_j
30	-480	31	-960	32	80
44	40	46	$-\frac{985544}{417}$	47	$-\frac{1276904}{139}$
48	$\frac{17615752}{5421}$	52	120	54	$-\frac{560744}{139}$
55	$-\frac{1753280}{139}$	56	$\frac{9541218}{1807}$	58	$-\frac{696688}{139}$
59	$-\frac{1545888}{139}$	60	$\frac{6447172}{1807}$	61	$-\frac{2298480}{139}$
62	$-\frac{31629680}{5421}$	63	$-\frac{18416956}{1807}$	64	$-\frac{12669404}{5421}$
76	-50	78	$\frac{224942}{417}$	79	$\frac{185462}{139}$
80	$\frac{743996}{5421}$	84	$-\frac{280}{3}$	86	$\frac{660508}{417}$
87	$\frac{346688}{139}$	88	$-\frac{434087}{2224}$	90	$\frac{72448}{139}$
91	$-\frac{322884}{139}$	92	$-\frac{9854045}{7228}$	93	$-\frac{1463144}{139}$
94	$\frac{7674219}{14456}$	95	$\frac{2933377}{834}$	96	$\frac{259435535}{86736}$
111	$\frac{258000}{139}$	112	$-\frac{3555872}{1807}$	116	80
118	$\frac{563662}{417}$	119	$-\frac{2868}{139}$	120	$-\frac{3082043}{6672}$
122	$-\frac{18272}{417}$	123	$-\frac{507776}{417}$	124	$-\frac{22787469}{14456}$
125	$-\frac{723812}{139}$	126	$-\frac{6757760}{5421}$	127	$-\frac{6947744}{5421}$
128	$-\frac{29083111}{86736}$	142	$-\frac{402598}{417}$	144	$-\frac{3716515}{1807}$
150	$\frac{157344}{139}$	151	$\frac{879564}{139}$	156	$\frac{185281999}{86736}$
158	$\frac{166192045}{86736}$	159	$\frac{2900207}{7228}$	160	$-\frac{5417647}{10842}$

B Reproduction and artifact contents

The standalone source is `article.tex`; the PDF is compiled from it without external figures or a bibliography database. The Python dependencies used in the delivered run are recorded in `requirements.txt` and `results/environment.json`. From the package directory, run

```
python verification/run_all.py
python build.py
```

The exact verifier does not need network access or a repository checkout. The numerical comparisons use mpmath. The complete coefficient certificates are in `certificates/`; the independent verifier accepts only restricted arithmetic syntax when reading them. The generated example table is `results/mixed_moments.json`. Original delivery evidence is in `results/`; rerunning a suite rewrites its corresponding result file, not repository source files.

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